Evaluated Under What? A Channel-Realism Audit of Semantic Communication for Non-Terrestrial Networks

Phuc Hao Do[a]

[a]*CAIRA-DAU, Center for Advanced Innovation, Research & Application, Danang Architecture University, Da Nang, Viet Nam*

Abstract

Semantic communication is recommended for non-terrestrial networks on the argument that it tolerates the conditions those links impose: large propagation delay, Doppler shifts of hundreds of kilohertz, short visibility windows and an elevation-dependent link budget. Surveys of the area map each of these limitations onto the works said to address it. We ask the narrower question those mappings leave open, namely whether the works cited as addressing a limitation are evaluated under it. Two measurements answer it. First, propagating 1,200 public Starlink two-line element sets with SGP4 over 15,356 distinct passes fixes the magnitude of each effect from orbital geometry rather than from assumption. At one station and a minimum elevation of 25°, over 1,978 of those passes, the peak Doppler shift of a pass has a median of 359.2 kHz at Ka-band and reaches 455.5 kHz, with a drift of up to 16.9 kHz/s, and the geometric term of the link budget sweeps 3.72 dB across a median visibility window of 206.0 s. Second, a coding of the 46 works a recent survey lists as representative, of which 27 could be obtained in full text, records for each whether an effect appears in the framing of the paper or in its evaluation. Of 26 codeable works, 16 evaluate over an additive white Gaussian noise channel, 4 use a standardised non-terrestrial channel model, 0 drive their channel from orbital data, and 0 instantiate the time-varying component of Doppler. Third, training a semantic codec under the modal assumption and evaluating it under the measured conditions locates the cost precisely. It is not where an audit would expect: sweeping the signal-to-noise ratio across the full pass costs 3.35 dB and degrades gracefully, so the fixed operating point most works adopt is defensible. The cost appears instead as a cliff on the axis nobody instantiates, with 17.96 dB lost at a residual frequency offset of 10^{-3} when that offset is allowed to accumulate across a coded block. What sets the cliff is the interval over which phase accumulates rather than Doppler as such: re-estimating carrier phase every 2 symbols removes the collapse entirely, and the cliff position scales inversely with the accumulation length across a factor of 256, at a constant 0.25 of the value the accumulated-phase argument predicts. The tolerable residual is therefore a property of the codec together with its carrier recovery, which is what no work in the population states. Randomising the offset during training recovers 13.77 dB of the open-loop loss. The checklist, the coding sheet with the sentence justifying every decision, the orbital generator and the evaluation harness are released.

Keywords: Semantic communication, Non-terrestrial networks, LEO satellites, Evaluation methodology, Reproducibility, Channel modelling

Email address: haodp.sut@gmail.com (Phuc Hao Do)

1. Introduction

Semantic communication proposes to transmit what a message means for a downstream task rather than the symbols that encode it, and it is increasingly recommended for non-terrestrial networks (NTN). The argument is one of fit. Satellite links impose constraints that a conventional pipeline handles poorly, and a recent survey of the area states them explicitly [1]: severe and time-varying path loss, long propagation delay, large and time-varying Doppler shift, intermittent connectivity with short visibility windows, atmospheric attenuation, on-board size, weight and power limits, and distortion that accumulates over multiple hops.

That survey does something more specific than list the constraints. It tabulates a mapping from each NTN limitation to the semantic-communication property said to address it, and cites representative works against each pairing. The mapping is a claim about the literature, and it is the claim this paper tests. A work can be cited as addressing Doppler because it discusses Doppler, and can nevertheless be evaluated over a channel in which no Doppler is present. Whether that gap exists, and how wide it is, has not been measured.

What this paper does. Figure 1 sets out the audit. Two measurements are taken separately and neither depends on the other. The first establishes what the orbit does, by propagating public two-line element sets and computing the quantities of Section 3 that an NTN-specific evaluation would have to instantiate. The second establishes what the literature evaluates, by coding the works a survey lists as representative against a checklist anchored to the 3GPP non-terrestrial channel studies [2] rather than to criteria of our own devising. A third measurement then asks what the gap costs, by training a semantic codec under the modal assumption of the population and evaluating it under the conditions the first measurement found.

What we do not claim. Three boundaries are set at the outset because an audit is easy to overstate. We do not claim the field is careless: the survey's own tables carry a channel-consideration column, so the community records this information, and our contribution is that nobody has compared the record against the evaluations. We propose no new semantic-quality metric, an area addressed by recent work that approaches evaluation from the metric side rather than the conditions side [3]. And we do not revisit whether learned codecs beat classical separation-based baselines, a question the deep joint source-channel coding literature already treats carefully [4, 5], including published cases where the classical baseline wins.

Two findings that run the other way. An audit is worth little if it only confirms its own premise, and two results here do not. The first is that the link budget moves slowly: Section 5 measures the sweep at 0.049 dB/s, which over ten seconds is 0.49 dB, so treating the channel as static within a transmission block is sound. The second is that even the pass-scale sweep is survivable: Section 7 finds a codec trained at one operating point loses 3.35 dB across the full range, degrading gradually and without a threshold. The commonest simplification in the population is therefore not the problem, and we report both because they narrow the audit to the one effect that does not survive measurement.

Where the cost actually is. That effect is the residual frequency offset, and more precisely the interval over which it is allowed to accumulate. Left to accumulate across a block, an offset of $\varepsilon = 10^{-3}$ costs 17.96 dB where $\varepsilon = 10^{-4}$ costs 1.17 dB, a cliff rather than a slope. Interrupting the accumulation moves it: re-estimating carrier phase every 2 symbols leaves the same codec

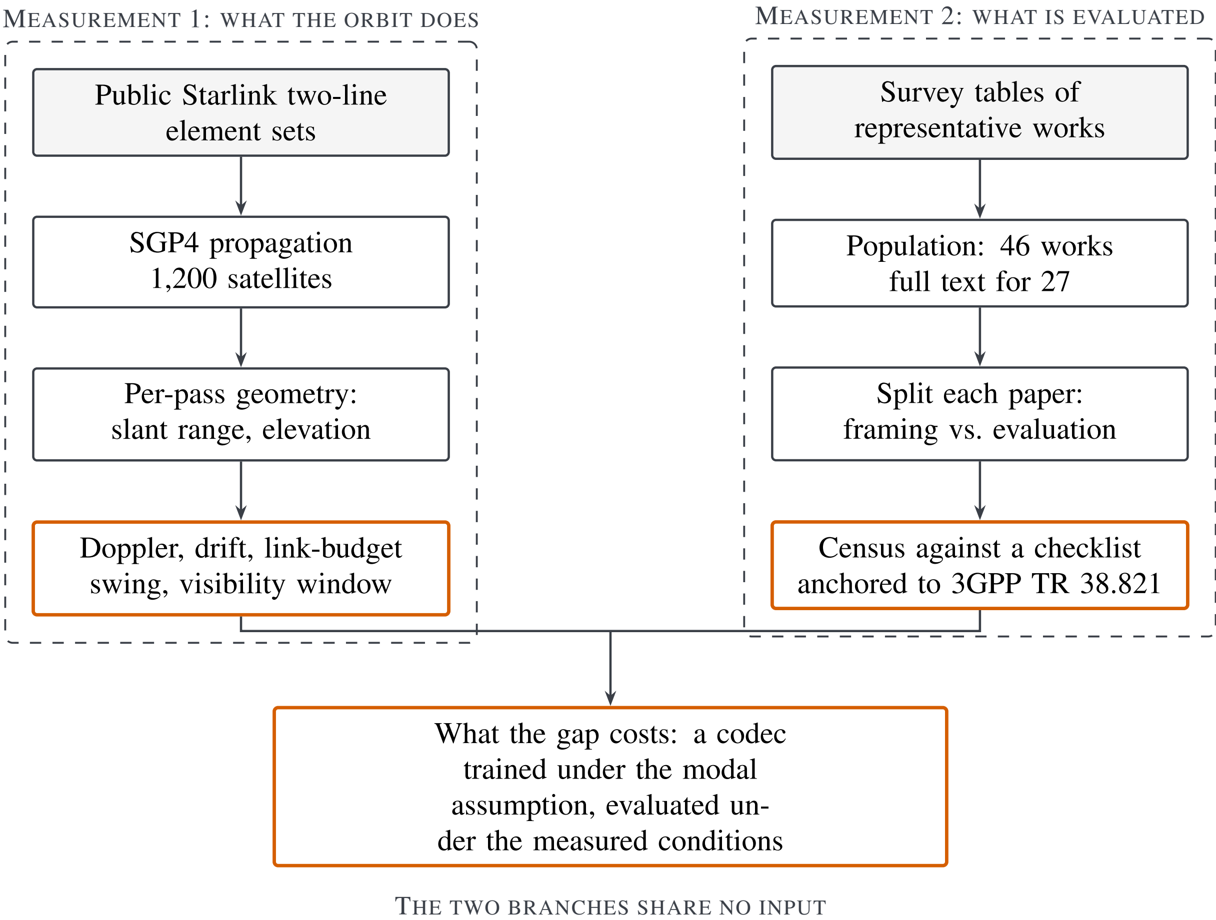

Figure 1: The audit. The two measurements are independent: the orbital side never consults the literature and the census never consults the orbit. They meet only in Section 7, where the conditions measured on the left are applied to a codec trained under the assumption counted on the right.

at 27.52 dB where the open-loop case has fallen to 10.08 dB, and the cliff position tracks the accumulation length across a factor of 256. What a deployed system tolerates is therefore set by the pair of codec and carrier recovery, and neither the residual nor the recovery interval is stated by the works in the population, of which 0 of 26 instantiate this axis at all.

2. Background

2.1. Why non-terrestrial links are said to be different

A satellite in low Earth orbit travels at roughly 7.5 km/s along its track, which is the origin of every constraint in the list above. What a ground station sees is the component of that motion along the line of sight, and it is the rate of change of the slant range (1) rather than the orbital speed itself that drives the effects below. Motion at that speed produces a Doppler shift proportional to the carrier frequency, so a link that is untroubled at S-band becomes demanding at Ka-band. It also bounds the time a given satellite is usable, since a pass ends when the elevation angle falls below the minimum the link budget supports. And because free-space path loss depends on slant range, which varies over the pass, the received power is not constant even in the absence of fading.

These quantities are geometric. They follow from orbital elements, a ground station position and a carrier frequency, and they can be computed rather than assumed. Section 5 does so, which is what allows the rest of the paper to speak about magnitudes instead of adjectives.

2.2. Semantic communication and its evaluation

Learned joint source-channel coding maps a source directly to channel symbols with a neural encoder and reconstructs it with a neural decoder, trained end to end through a differentiable channel [4]. The shift of objective from reproducing symbols to preserving meaning is a departure from the classical formulation [6] and is set out in [7]. The channel is therefore not an external assumption but a component of the training objective, which is why the choice of channel model has consequences that reach beyond the reported numbers: a codec learns to invert the impairments it is trained against and has no reason to handle those it never saw.

Evaluation practice in the area has begun to receive attention from the metric side, asking how semantic success should be defined and measured across modalities and tasks [3]. The present paper is complementary and asks about conditions rather than metrics: given any metric, under what channel was it computed.

2.3. Semantic communication over non-terrestrial links

The population audited in this paper spans three settings, and it is worth describing them because the audit's question applies differently to each.

Satellite links. Earth observation is the dominant application, since the downlink is the bottleneck and the receiver's task is usually known in advance. Work here ranges from resource-constrained network optimisation for observation downlinks [8] and on-board compression for platforms with limited payload compute [9], through cognitive augmentation of observation constellations [10], to importance-aware transmission that protects the semantically salient part of an image [11] and generative reconstruction at the receiver [12, 13]. Multi-hop settings introduce inter-satellite routing as a further degree of freedom [14, 15], and direct satellite-to-device links remove the gateway altogether [16, 17].

Aerial platforms. Unmanned aerial vehicles share the bandwidth pressure but not the orbital geometry: their relative velocity is orders of magnitude lower, so Doppler is not the binding constraint, while energy and trajectory are. Work here couples semantic transmission with scheduling and trajectory [18, 19], with knowledge-graph representations for task-specific transfer [20, 21], and with resource allocation under fairness constraints [22, 23]. This paper reports the census over aerial work alongside the satellite work, but the checklist items that concern orbital motion are correspondingly less demanding there, and Section 9 returns to the point.

Integrated space-air-ground networks. A third group treats the tiers jointly, with federated or cooperative learning spanning them [24, 25], digital twins of the integrated network [26], aerial relays between tiers [27], and resource allocation that mixes conventional and semantic traffic [28, 29, 30]. Relaying frameworks that forward semantic representations rather than symbols [31] and goal-oriented formulations aimed at links where round-trip delay dominates [32] sit alongside them.

What none of this work is asked here is whether it is correct or useful. The question is narrower and mechanical: under which channel was the reported number computed.

2.4. Channel models and where the standards sit

The 3GPP non-terrestrial studies define reference scenarios, carrier bands and calibration assumptions for exactly the links these works target [33, 2]. They are the natural anchor for a checklist about channel realism, because they are external to the semantic communication literature and were written for a different purpose, so a work cannot be accused of being measured against criteria invented to catch it. Their reference bands, S-band and Ka-band, are the two this paper reports, and their minimum elevation angles are the thresholds swept in Section 5.

2.5. Reassessment as a contribution

Re-running a body of work under stricter conditions has an established place in networking, and the contribution in such papers is the protocol rather than a new mechanism. Reported gains in encrypted traffic classification have been traced to dataset-specific shortcuts rather than to the task semantics the models were credited with learning [34]. In adjacent areas, reality checks have been published even where the reassessed method loses to a conventional baseline, because what the community gains is a fair comparison protocol [35]. This paper is of that kind, and it is deliberately narrow: it does not evaluate the quality of any method, only the conditions under which quality was measured.

3. System model

This section fixes the quantities an evaluation of semantic communication over a non-terrestrial link would have to instantiate. Each is a function of orbital geometry and carrier frequency alone, so each can be computed rather than assumed, and Section 5 computes them.

3.1. Link geometry

Let R_E denote the Earth radius, h the orbital altitude of a satellite and $\varepsilon_{\mathrm{el}}$ the elevation angle at which a ground station sees it. The slant range follows from the triangle formed by the station, the satellite and the Earth centre,

$$d(\varepsilon_{\mathrm{el}}) \;=\; \sqrt{R_E^2 \sin^2 \varepsilon_{\mathrm{el}} + h\,(2R_E + h)} \;-\; R_E \sin \varepsilon_{\mathrm{el}}, \tag{1}$$

and the corresponding Earth-central angle, which fixes the ground footprint, is

$$\gamma(\varepsilon_{\mathrm{el}}) \;=\; \arccos\!\Big(\tfrac{R_E}{R_E+h}\,\cos\varepsilon_{\mathrm{el}}\Big) \;-\; \varepsilon_{\mathrm{el}}. \tag{2}$$

A pass is the interval during which $\varepsilon_{\mathrm{el}}(t) \geq \varepsilon_{\min}$ for a minimum elevation $\varepsilon_{\min}$ set by the link budget, and its duration T_{vis} is the visibility window. Equations (1) and (2) are monotone in $\varepsilon_{\mathrm{el}}$, so raising $\varepsilon_{\min}$ shortens T_{vis} and reduces the extremes of every quantity below; Section 5 reports that trade at three thresholds.

3.2. What varies within a pass

Path loss. The free-space term at carrier frequency f_c is

$$L(d, f_c) \;=\; 20\log_{10} d \;+\; 20\log_{10} f_c \;+\; 92.45, \tag{3}$$

with d in kilometres and f_c in gigahertz. The quantity that matters for an evaluation is not L itself, which a link budget absorbs, but how much it moves across a pass:

$$\Delta L \;=\; 20\log_{10}\frac{d_{\max}}{d_{\min}} \;=\; 20\log_{10}\frac{d(\varepsilon_{\min})}{d(\varepsilon_{\mathrm{el}}^{\star})}, \tag{4}$$

where $\varepsilon_{\mathrm{el}}^{\star}$ is the maximum elevation reached. The frequency term of (3) cancels in (4), so the swing is a property of the geometry alone and is identical at S-band and at Ka-band.

Rate of change. Differentiating (3) gives the slew rate of the geometric term,

$$\frac{\mathrm{d}L}{\mathrm{d}t} \;=\; \frac{20}{\ln 10}\,\frac{\dot{d}}{d}, \tag{5}$$

so a transmission of duration T_{blk} sees a change bounded by

$$|\Delta L_{\mathrm{blk}}| \;\le\; T_{\mathrm{blk}}\cdot \max_t\left|\frac{\mathrm{d}L}{\mathrm{d}t}\right|. \tag{6}$$

Equations (4) and (6) are the two halves of the result in Section 5 that runs against the audit: the first is large and the second is not.

Doppler. The shift and its rate are

$$f_{\mathrm{dop}}(t) \;=\; -\frac{\dot{d}(t)}{c}\,f_c, \qquad \dot{f}_{\mathrm{dop}}(t) \;=\; -\frac{\ddot{d}(t)}{c}\,f_c, \tag{7}$$

with c the speed of light. Both scale linearly with f_c, which is why a link that is untroubled at S-band is demanding at Ka-band, and $\dot{d}$ changes sign at closest approach, so f_{dop} reverses sign within a single pass. The one-way delay is d/c.

What the model omits, and in which direction. Equation (3) is the free-space term only. Antenna pattern roll-off as the beam sweeps across the station, atmospheric and rain attenuation, which grow as $\varepsilon_{\mathrm{el}}$ falls, and receiver noise all add variation over a pass. Every one of them increases the right-hand side of (4), so the figures reported in Section 5 are lower bounds. A lower bound requires no contested modelling and is all the argument of this paper needs.

3.3. Semantic transmission over such a link

A learned joint source-channel codec maps a source $\mathbf{x}\in\mathbb{R}^n$ to channel symbols and back,

$$\mathbf{z} = f_\theta(\mathbf{x}) \in \mathbb{R}^{2k}, \qquad \hat{\mathbf{x}} = g_\phi(\mathbf{y}), \tag{8}$$

where consecutive pairs of $\mathbf{z}$ form k complex symbols $\mathbf{s}\in\mathbb{C}^k$ and $R = k/n$ is the bandwidth ratio. Transmit power is normalised per source block,

$$\tilde{\mathbf{s}} \;=\; \sqrt{k}\,\frac{\mathbf{s}}{\|\mathbf{s}\|_2}, \tag{9}$$

so that $\mathbb{E}\,|\tilde{s}_m|^2 = 1$ and the channel signal-to-noise ratio is $\gamma = 1/\sigma^2$.

The channel the population assumes, and the one the orbit provides. The modal choice in the population of Section 6 is

$$y_m \;=\; \tilde{s}_m + n_m, \qquad n_m \sim \mathcal{CN}(0,\sigma^2), \tag{10}$$

evaluated at a single γ. Adding the residual frequency offset that (7) implies gives

$$y_m \;=\; \tilde{s}_m\, e^{j2\pi\varepsilon m} + n_m, \qquad \varepsilon \;=\; \frac{f_{\mathrm{res}}}{R_s} \;=\; \frac{\rho\, f_{\mathrm{dop}}}{R_s}, \tag{11}$$

where R_s is the symbol rate and $\rho \in [0,1]$ is the fraction of the shift that compensation leaves uncorrected. Note that ε in (11) is *constant* within a block, so the phase it produces grows linearly with m. That is a fixed frequency offset, not the time-varying Doppler of the checklist, which is a non-zero $\dot{f}_{\mathrm{dop}}$ in (7) and would appear as a quadratic phase. The model of this paper therefore does not itself instantiate the checklist item it finds nobody instantiating, and does not claim to. Parameterising by the normalised offset ε rather than by f_{res} avoids committing to a symbol rate, which no work in the population states. The phase error is not a random perturbation but accumulates deterministically across the block,

$$\Delta\varphi \;=\; 2\pi\varepsilon(k-1), \tag{12}$$

so the last symbols of a block are rotated far more than the first. Setting $\Delta\varphi = \pi$ in (12) gives the offset at which the tail of a block is inverted, $\varepsilon = 1/(2(k-1))$, which for the operating point of Section 7 is close to 10^{-3} and is where the collapse reported there occurs.

Fidelity. Reconstruction quality is reported as

$$\mathrm{PSNR} \;=\; 10\log_{10}\frac{M^2}{\frac{1}{n}\|\mathbf{x}-\hat{\mathbf{x}}\|_2^2}, \tag{13}$$

with M the peak source value, and the codec is trained by minimising the mean squared error in the denominator of (13) through (10).

4. The audit protocol

4.1. Population

The population is the set of works a recent survey of semantic communication for non-terrestrial networks [1] lists in its representative-works tables, covering satellite, aerial and integrated scenarios. Using a survey as the sampling frame is a convenience sample and we treat it as one, with the consequences discussed in Section 9. It has one property that outweighs the cost: the pairing between an NTN limitation and the works said to address it is made by the field rather than by us, so the audit tests a claim the community has already put in print.

The tables list 46 distinct works [9, 11, 10, 8, 28, 15, 16, 12, 21, 36, 31, 37, 17, 24, 30, 26, 25, 23, 22, 20, 18, 32, 27, 14, 13, 29, 19]. Full text could be obtained for 27, which is 58.7% of the population; the remaining 19 are counted as not auditable rather than guessed at, since a work whose evaluation section cannot be read is a fact about the accessibility of the literature and not a gap to be filled by inference.

One further work, reference [108] of the source survey, is retrieved but not coded. It is a position paper on goal-oriented communication for interplanetary and non-terrestrial links and carries no conventional evaluation section, so the partition of Section 4.4 cannot be applied to it. Table A.7 in the appendix enumerates all 46 works, giving for each the reference number the source survey assigns it, the entry in this paper's bibliography where one exists, its title, and whether it was retrieved and coded. The 19 works that could not be retrieved are named there rather than left as a count, and the table is the crosswalk that makes the survey reference numbers used throughout Section 6 resolvable. Excluding the position paper leaves the 26 works on which every count in Section 6 is computed. Of those, 15 come from the satellite tables, 7 from the aerial tables and 4 from the integrated tables, a split that Section 9 uses to check the main result.

4.2. Checklist

Table 1 states the checklist: 12 items covering the physical effects an NTN evaluation could instantiate and the channel models it could use, each with the source it is anchored to. 7 of the items restate structural constraints the field's own survey names, the standardised-model item points at the 3GPP studies, and the remainder are protocol conventions rather than physical effects. Two design decisions matter. The effects are those the field's own survey names as structural NTN constraints, and the channel-model entries follow the 3GPP non-terrestrial studies [33, 2], so neither list is ours. And the checklist is deliberately about *conditions of evaluation*, not about quality: a work that evaluates over additive white Gaussian noise is not thereby wrong, it is evaluated under a stated condition, and the audit records which condition.

4.3. Coding, and where a claim counts

The instrument, stated formally in Section 4.4, is a distinction between two regions of a paper. A sentence about Doppler in an introduction or a related-work section states a motivation. The same sentence in a simulation setup or results section is evidence that the evaluation instantiates the effect. Every paper is therefore split at the first heading that begins its evaluation, and every checklist item is coded separately for the region before and the region after.

The extraction step returns sentences rather than verdicts. For each work and each checklist item, the harness collects the sentences that bear on the item, tagged by region, and a decision is recorded against those sentences. The coding sheet released with this paper carries the sentence behind every decision, so any individual judgement can be disputed without repeating the collection.

4.4. The census, stated precisely

Let $\mathcal{W}$ be the population and $\mathcal{I}$ the checklist of Table 1, with $|\mathcal{I}| = 12$. Each work w is partitioned at the first heading that begins its evaluation into a framing region and an evaluation region, and the coding function

$$c(w,i) \;=\; \begin{cases} 2, & \text{item } i \text{ is evidenced in the evaluation region of } w,\\ 1, & \text{evidenced only in the framing region,} \\ 0, & \text{not evidenced,} \end{cases} \tag{14}$$

records where each item appears. The distinction in (14) between 2 and 1 is the instrument of this paper: a statement about Doppler while motivating a problem is not evidence that the evaluation

Table 1: The checklist, with the source each item is anchored to. "Survey" items restate structural constraints named by the field's own survey, "3GPP" points at the non-terrestrial channel studies, "Orbit" follows from orbital mechanics, and "Protocol" items are evaluation conventions rather than physical effects.

Item	Anchor	What the item asks the evaluation to instantiate
Doppler shift	Survey	Large and time-varying Doppler shift
Time-varying Doppler	Survey	the time-varying component of the same constraint
Propagation delay	Survey	Long propagation delay
Elevation-dependent loss	Survey	Severe and time-varying path loss
Finite visibility window	Survey	Intermittent connectivity, short visibility windows
Atmospheric attenuation	Survey	Atmospheric and weather attenuation
On-board compute limit	Survey	On-board size, weight and power limits
AWGN channel	Protocol	conventional terrestrial reference channel
Statistical fading	Protocol	conventional statistical fading channel
Standardised NTN model	3GPP	TR 38.811 / TR 38.821 non-terrestrial channel
Real orbital trace	Orbit	channel driven by propagated orbital elements
SNR sweep / mismatch	Protocol	operating range rather than operating point

instantiates Doppler. Writing $\mathcal{W}_c \subseteq \mathcal{W}$ for the works that could be read and partitioned, the per-item count reported in Section 6 is

$$n_{\text{ev}}(i) \;=\; \big|\{\, w \in \mathcal{W}_c : c(w,i) = 2 \,\}\big|, \tag{15}$$

and the per-work coverage reported there is

$$\kappa(w) \;=\; \big|\{\, i \in \mathcal{I} : c(w,i) = 2 \,\}\big|. \tag{16}$$

Comparing two descriptions of the same work. The survey's channel-consideration entry is a second, independent description of what a work considers. Let $\mathcal{A}(w) \subseteq \mathcal{I}$ be the items that entry names and

$$\mathcal{B}(w) \;=\; \{\, i \in \mathcal{I} : c(w,i) = 2 \,\} \tag{17}$$

the items its evaluation instantiates. Because an entry of a few words cannot enumerate an evaluation, $\mathcal{A}(w) \subseteq \mathcal{B}(w)$ is the expected relation and carries no information. The informative quantity is the discrepancy set

$$\mathcal{D}(w) \;=\; \mathcal{A}(w) \setminus \mathcal{B}(w), \tag{18}$$

the effects a work is recorded as considering but does not instantiate where it matters. Section 6 reports $|\mathcal{A}(w)|$, $|\mathcal{B}(w)|$ and $\mathcal{D}(w)$ for every work where both descriptions could be read.

Agreement between coders. Where two coders assign (14) independently, agreement is reported as

$$\kappa_{\mathrm{C}} = \frac{p_o - p_e}{1 - p_e}, \tag{19}$$

with p_o the observed proportion of matching decisions and p_e the proportion expected by chance. **No value of (19) is reported here, because a second coder was not available.** The procedure is stated so that the estimate can be supplied by anyone extending this work, and the released coding sheet carries the sentence behind every decision, so a disputed entry can be settled against the evidence rather than against a coefficient.

That substitution is adequate for items where evidence exists and a coder could disagree about it. It is not adequate for the two items that return zero, where there is no sentence for two coders to read and disagree over, and where a coefficient computed over the other items would say nothing about them. The zeros are therefore defended differently, by asking whether the detector was too narrow rather than whether a second reader would agree with it, and Section 6.1 does that.

4.5. Testing the instrument before trusting it

Two checklist items return zero across the whole population, and a zero is exactly where a broken detector is indistinguishable from a finding. Three controls therefore run before any zero is quoted. Sentences written the way papers actually write them must be caught, including phrasings that avoid the obvious keyword. Near misses must be rejected, among them a mention of 3GPP that is not a channel model. And the detectors must demonstrably fire on the real corpus, which they do, returning 262 matching sentences across the 27 retrieved works, ruling out an empty-input failure in which every count is zero because nothing was read. All three pass, and the controls are released as an executable script rather than described.

5. What the orbit actually does

5.1. Method

A seeded sample of 1,200 Starlink two-line element sets is propagated with SGP4 [38] over 15,356 distinct passes at four ground stations spanning 0 to 65 degrees of latitude. For each pass we record the slant range of (1), the elevation, the one-way delay, the Doppler shift and its rate from (7) at S-band and at Ka-band, the free-space term of (3), its swing (4) and its slew rate (5).

A pass is counted once here even though it is analysed at three elevation thresholds. Counting each threshold separately would give 30,922 pass-threshold observations, but a pass that clears 30° also clears 25° and 10°, so that figure counts the same contact up to three times. The distinct count is the one at the loosest threshold, 10°, since the stricter sets are subsets of it.

Only geometry, deliberately. As set out after (4), we compute the free-space term and nothing else, so every figure below is a lower bound on the variation a deployed link would see.

5.2. Results

Figure 2 shows the three distributions that matter and Table 2 reports them per station. Peak Doppler at Ka-band has a median of 359.2 kHz and reaches 455.5 kHz, which is the literal content of the phrase "hundreds of kilohertz" that motivates the area. At S-band the same passes give 35.9 kHz. The drift has a median of 5.4 kHz/s and a maximum of 16.9 kHz/s, so the shift is not a constant offset that a single estimate removes. The geometric term of the link budget sweeps a median of 3.72 dB and up to 6.72 dB over a median visibility window of 206.0 s, and one-way delay reaches 3.25 ms.

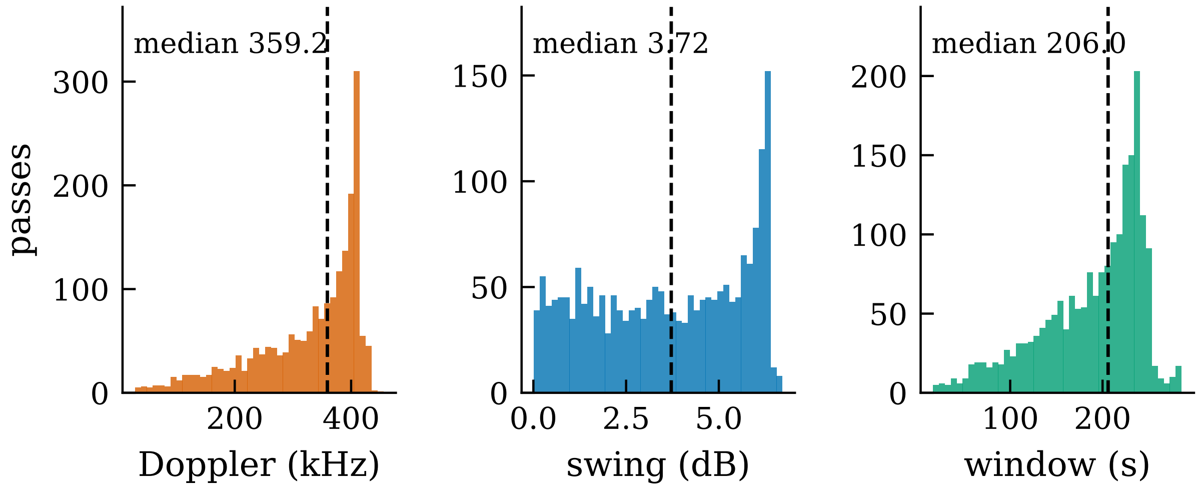

Figure 2: Distributions over 1,978 passes at one station with a minimum elevation of 25°: peak Doppler shift at Ka-band, the swing of the geometric link-budget term across a pass, and the visibility window. Dashed lines mark medians. The spread of the centre panel is explained by geometry rather than assumed to be: swing correlates with maximum elevation at $r = 0.962$, and the third of passes that stay lowest sweep a median of 1.20 dB against 6.02 dB for the third that pass highest.

Table 2: Medians per ground station at a minimum elevation of 25°. The quantities differ by at most 15.6 kHz in Doppler across 65 degrees of latitude, so they are properties of the orbital shell rather than of a chosen site.

Ground station	Lat. (°)	Passes	Window (s)	Swing (dB)	Doppler (kHz)
Da Nang	16.0	1,978	206.0	3.72	359.2
Equator	0.0	1,878	205.5	3.67	356.3
Mid-lat	45.0	3,663	200.5	3.61	354.5
High-lat	65.0	927	215.0	3.87	370.1

5.3. One pass, in detail

Figure 3 follows a single pass, and the pass is chosen by a stated rule rather than for appearance: it is the one whose maximum elevation lies closest to the median of the distribution. Over its 216.0 s the Ka-band shift runs from -352.4 kHz to 353.1 kHz, an excursion of 705.5 kHz that passes through zero at closest approach, so the sign of the offset reverses within a single contact. The shaded band marks a one-second transmission block at the same scale. Its width against the pass is the visual form of the two results of this section: nothing of consequence happens to the link budget inside the block, and a great deal happens across the pass that contains it.

5.4. Sensitivity to the elevation threshold

The minimum elevation a link supports is a design choice, and it moves every quantity above. Figure 4 and Table 3 report the three thresholds the 3GPP studies use. Raising the threshold from 10° to 30° cuts the number of usable passes at one station from 3,662 to 1,672 per day while shortening the window, so a stricter threshold buys a better link at the cost of contact time. The Doppler extremes fall because the geometry excludes the low-elevation part of the pass where the range rate is largest, but they remain in the hundreds of kilohertz at every threshold.

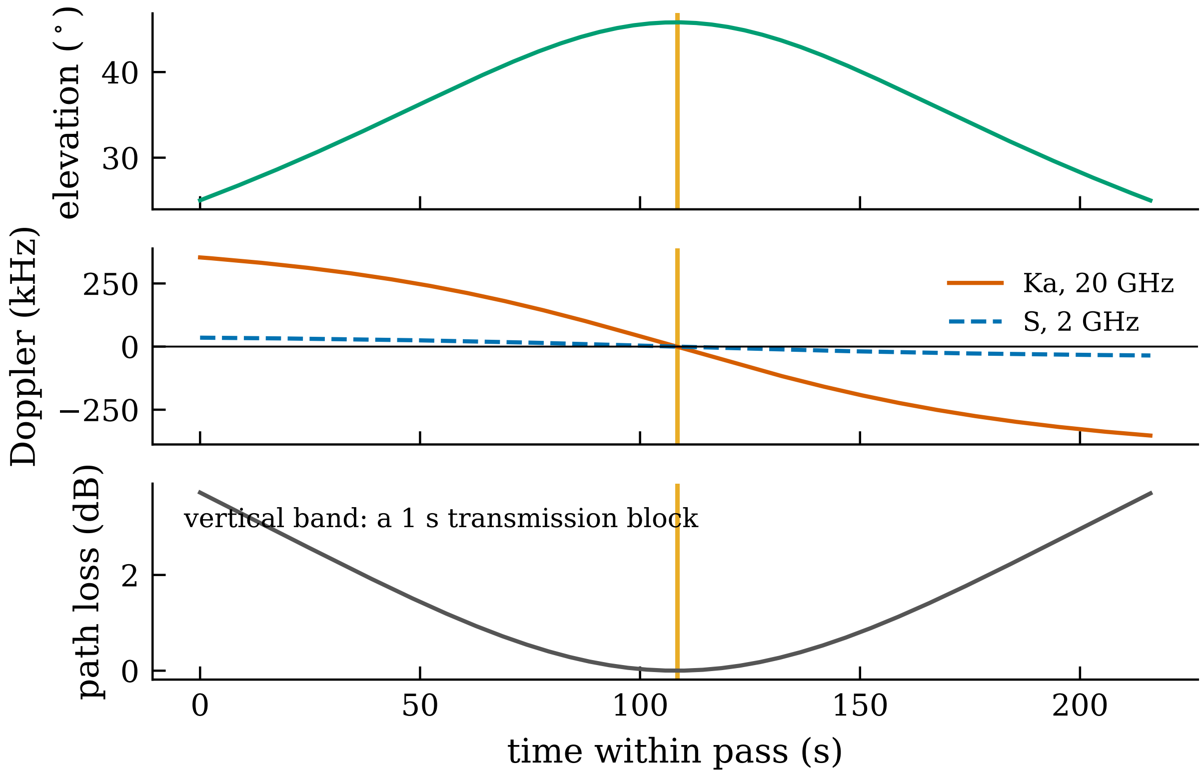

Figure 3: One pass of Starlink 5768 over the station, reaching 45.8°. The selection rule is stated in the text and in the released code. Path loss is plotted relative to its minimum.

5.5. The result that runs against the audit

The slew rate (5) has a median of 0.049 dB/s, so by (6) a transmission block sees very little. Over a one-second transmission the signal-to-noise ratio therefore shifts by well under a tenth of a decibel, and over ten seconds by 0.49 dB. The common practice of holding the signal-to-noise ratio fixed within a transmission block is sound, and an audit that attacked it would be attacking the wrong thing.

What this leaves is a different question, and it is the one Section 7 asks. The sweep is small per second but not small per pass. A codec trained at one operating point is deployed across a window of 206.0 s over which the geometric term alone moves 3.72 dB.

6. What the literature actually evaluates

Figure 5 and Table 4 report $n_{\mathrm{ev}}(i)$ of (15) over the $|\mathcal{W}_c| = 26$ works for which full text was available and an evaluation section could be located, alongside the corresponding count over framing regions.

Channel models. Additive white Gaussian noise is the modal choice, present in the evaluation of 16 works. Statistical fading appears in 8. A standardised non-terrestrial channel model appears in 4. No work in the population drives its channel from orbital data: the count for two-line elements, SGP4, ephemeris or an equivalent propagation tool is 0.

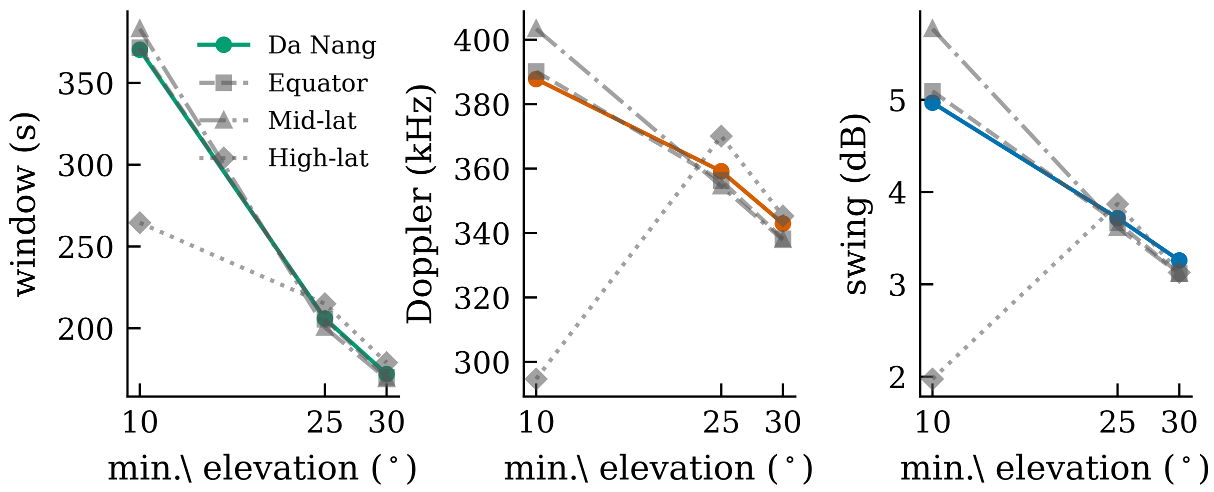

Figure 4: Medians of the same three quantities, visibility window, peak Ka-band Doppler and link-budget swing, against the minimum elevation threshold, for four stations. The station used elsewhere in this paper is drawn in colour and the others in grey; the curves lie close together, which is the same shell property Table 2 reports.

Table 3: Full statistics at one station for the three elevation thresholds. Link-budget figures are the geometric term only and are therefore lower bounds. Two entries look contradictory and are not. Peak Doppler falls as the threshold rises because the largest range rate occurs near the horizon, which a stricter threshold discards, while median drift rises because the largest drift occurs at closest approach, which a stricter threshold retains. The maximum drift is identical at all three thresholds because the single most extreme pass clears 30° and so belongs to every set.

	10°		25°		30°	
Quantity	median	max	median	max	median	max
Visibility window (s)	370.2	502.0	206.0	285.5	172.0	241.0
One-way delay (ms)	5.41	6.26	3.25	3.90	2.86	3.45
Peak Doppler, S-band (kHz)	38.8	49.7	35.9	45.6	34.3	43.4
Peak Doppler, Ka-band (kHz)	387.8	497.0	359.2	455.5	342.9	433.5
Doppler drift, Ka (kHz/s)	3.66	16.88	5.36	16.88	5.73	16.88
Link-budget swing (dB)	4.97	12.15	3.72	6.72	3.26	5.45
Link-budget slew (dB/s)	0.034	0.143	0.049	0.143	0.052	0.143

Effects. Doppler appears in the evaluation of 11 works and in the framing of 2. Its time-varying component appears in 0. Elevation-dependent loss appears in 9, propagation delay in 8, a finite visibility window in 3, and an on-board compute limit in 3.

6.1. Were the detectors too narrow? A sensitivity test on the zeros

A zero invites one specific objection: that the vocabulary searched for was too narrow, and that the effect is present under wording the detector never looked for. The controls of Section 4 answer the opposite question, whether the detector fires when it should, so they do not settle this one.

We therefore re-scanned the corpus for the two zero items with a deliberately wide net, several times the length of the original pattern and including phrasings that avoid the obvious keyword: for time-varying Doppler, terms such as frequency drift, frequency ramp, phase acceleration, second-order or quadratic phase, chirp and linear frequency modulation; for orbital traces, terms such as ephemeris, Keplerian, orbital element, Walker constellation, orbit propagation, Celestrak and Space-Track, together with the names of the common propagation libraries and of Systems Tool Kit.

Across 8,914 sentences in 27 works the wide net returns 1 sentence for time-varying Doppler and 1 for orbital traces, and neither is an evaluation condition. The first attributes a frequency drift to *user mobility* on an aerial link while motivating the problem. The second lists orbital elements among the things a digital twin should represent semantically, which is a statement about what to encode rather than about how a channel was generated. Both are released with the scan so the judgement can be checked.

A pattern several times longer than the original, applied to 8,914 sentences, therefore adds 1 and 1 candidate sentences and no evaluation condition: the zeros are not an artefact of a narrow vocabulary. What they remain vulnerable to is a work that instantiates the effect without describing it in any of these terms, which Section 9 states as a residual threat.

Reading the two zeros. The zeros for time-varying Doppler and for orbital traces are measurements, not silence: the controls of Section 4 establish that the detectors catch the relevant phrasings and do fire elsewhere in the same corpus. Read against Section 5, they say that the drift the orbit produces, up to 16.9 kHz/s, is instantiated by no evaluation in the population, and that no evaluation is driven by the geometry that produces it.

The structure behind the counts. Figure 6 resolves the census to individual works. The coverage $\kappa(w)$ of (16) never exceeds 7 of the 12 items for any work in the population, and 2 works have $\kappa(w) = 0$. The two empty rows are the zeros discussed above, and seeing them as rows rather than as bars makes clear that they are not the result of one or two works dominating a count.

A second source describes the same works. The survey's own channel-consideration column is an independent record of what each work considers, so it can be compared against what the evaluation sections contain. Section 6 reports $|\mathcal{A}(w)|$, the set $\mathcal{B}(w)$ of (17) and the discrepancy $\mathcal{D}(w)$ of (18) for the 15 works where both descriptions are available. Two things follow, and the first is not a criticism. The column is a phrase of a few words and cannot enumerate an evaluation, so the entries naming fewer items than the evaluations contain is expected. Table 5 lists them. What is informative is what the entries name at all: across 15 works they name 2 checklist items in total, because most entries describe an architecture or a topology rather than a propagation environment. The column is the closest thing the literature has to a record of channel realism, and it is not one. In 1 case $\mathcal{D}(w)$ is non-empty, with the column naming an effect the evaluation section does not contain.

7. What the gap costs

The census establishes what the population evaluates and Section 5 establishes what the orbit does. This section joins them by taking the modal assumption of the population, an additive white Gaussian noise channel at a single signal-to-noise ratio, and asking what happens to a codec trained under it when the conditions of Section 5 are applied.

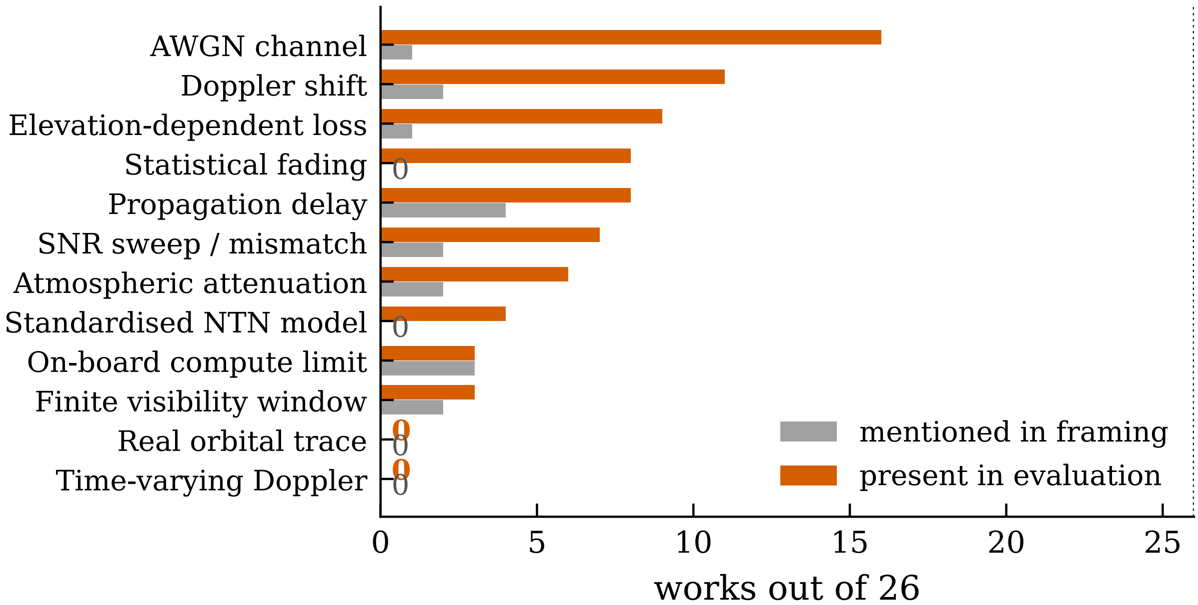

Figure 5: Where each effect appears, for the 26 codeable works. Grey counts works that mention the effect only while motivating the problem; orange counts works whose evaluation instantiates it. Zeros are labelled explicitly so that a measured zero is not read as a missing bar.

Table 4: Census over 26 works. "Framing" counts appearances before the evaluation section, "Evaluation" counts appearances within it. A work may appear in both columns.

Code	NTN effect or channel model	Framing	Evaluation
`CH_AWGN`	AWGN channel	1	16
`DOPPLER`	Doppler shift	2	11
`PATHLOSS_ELEV`	Elevation-dependent loss	1	9
`CH_FADING`	Statistical fading	0	8
`DELAY`	Propagation delay	4	8
`SNR_SWEEP`	SNR sweep / mismatch	2	7
`ATMOS`	Atmospheric attenuation	2	6
`CH_3GPP`	Standardised NTN model	0	4
`ONBOARD`	On-board compute limit	3	3
`VISIBILITY`	Finite visibility window	2	3
`CH_TRACE`	Real orbital trace	0	0
`DOPPLER_RATE`	Time-varying Doppler	0	0

7.1. Setup

The codec of (8) and (9) is instantiated as a convolutional encoder and decoder [4] and trained end to end through the differentiable channel (10) on a standard image benchmark [39], with fidelity measured by (13). Two training regimes are compared. The first fixes the signal-to-noise ratio during training, which is the modal practice in the population. The second randomises it, together with a residual frequency offset, over the ranges Section 5 measured. The second regime

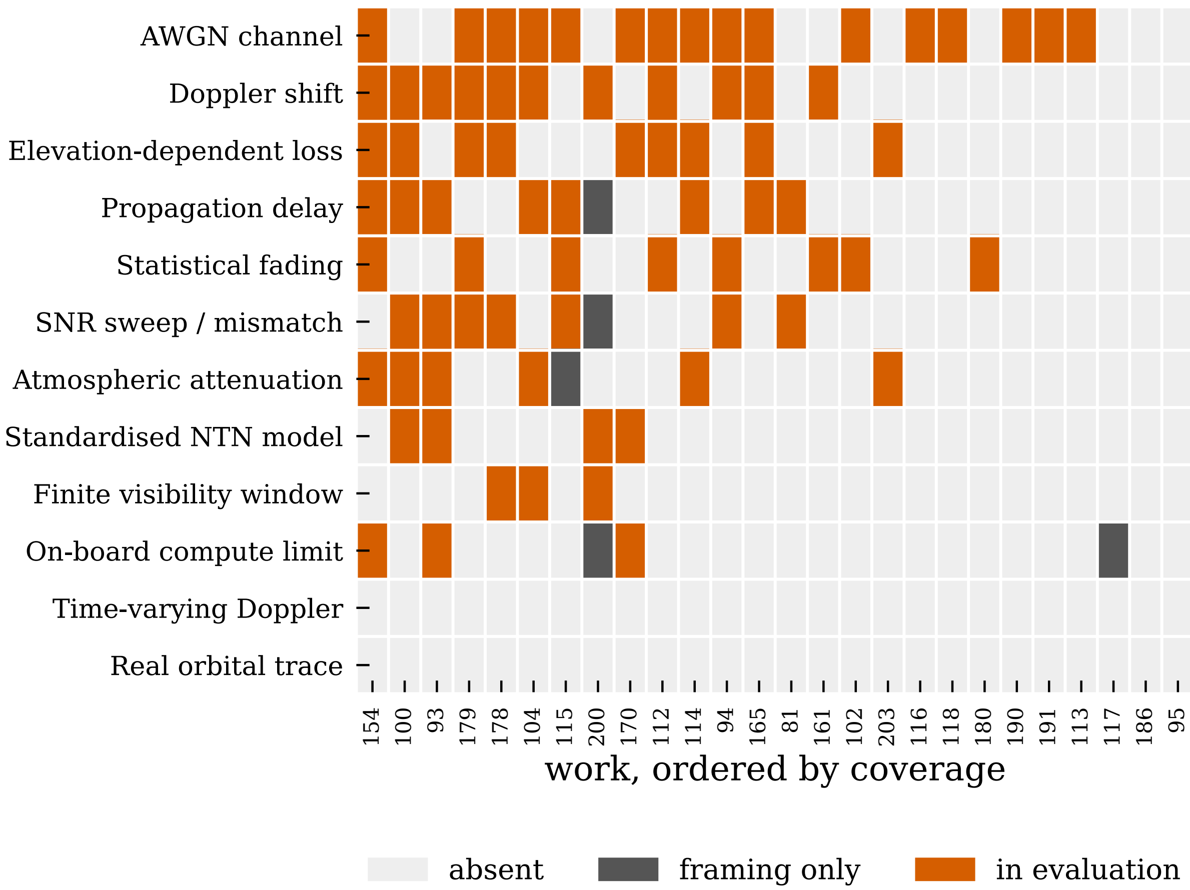

Figure 6: Every codeable work against every checklist item. Columns are labelled by the reference number the source survey gives the work and ordered by how many items its evaluation instantiates. Grey marks an effect that appears only while the problem is being motivated; orange marks one the evaluation instantiates.

is included so that the paper reports a remedy rather than only a deficiency, and so that the cost of the gap can be separated from the cost of fixing it.

The evaluation range is not chosen by us. The sweep spans the link-budget swing measured in Section 5 rather than a range picked for effect, and because that swing excludes antenna pattern and atmospherics it is the conservative end of what a deployed codec would see.

Residual offset, not raw Doppler. No deployed system leaves a shift of order 359.2 kHz uncompensated, so evaluating at raw Doppler would be a straw man. We parameterise instead by the normalised residual offset ε of (11), which avoids committing to a symbol rate that no work in the population states, and whose effect accumulates across a block as (12) describes. A reader substitutes their own R_s to locate their operating point: at the median Ka-band shift of 359.2 kHz and a symbol rate of 10 Msym/s, leaving one per cent of the shift uncorrected puts ε near 3.6×10^{-4}.

Table 5: The survey's channel-consideration entry against the evaluation sections, for the 15 works where both could be read. "Named" counts checklist items the entry mentions, "In eval." counts items the evaluation instantiates.

Work	Survey's channel entry	Named	In eval.	Named but absent
[95]	EO downlink, visibility window	1	0	Finite visibility window
[81]	Multi-hop ISL routing	0	2	—
[93]	Adaptive encoder-decoder	0	6	—
[94]	SNR-aware codec	0	4	—
[100]	Doppler, weather fading	1	6	—
[104]	Hierarchical FL	0	5	—
[113]	On-board sparse encoder	0	1	—
[115]	Multi-gateway hops, SNR-aware	0	4	—
[116]	Cooperative relaying	0	1	—
[117]	Semantic forwarding	0	0	—
[118]	Task-clustered FL	0	1	—
[154]	Inter-satellite links	0	7	—
[161]	Satellite interference	0	2	—
[165]	Direct Satellite Communication	0	4	—
[170]	SNR-aware link	0	4	—

7.2. Configuration, and which runs count

The codec of (8) uses a bandwidth ratio of $R = 0.167$, giving 512 complex symbols per source block, and each configuration is trained for 30 epochs with 13 seeds.

A run can fail, and a failed run is not a measurement. Training this architecture is not always stable: a run can converge and then diverge to a constant output, which the harness will happily report as a low reconstruction quality. Since a low number is exactly what the hypothesis under test predicts, such a run would be a failure that looks like a confirmation. We therefore declared a validity criterion before running, namely that a run counts only if its quality on the *clean* channel exceeds 20.0 dB, and we report the outcome rather than quietly dropping the failures. Of 18 runs, 12 were valid and 6 diverged, and every figure below uses the valid runs and states the count: 7 for the fixed regime and 5 for the randomised one. Gradient clipping reduced but did not eliminate the failures, so the criterion does work that the recipe alone does not.

Two properties of that failure are worth recording because they bear on reproduction. It is not seed-determined: the same seed produced a valid run in one execution and a divergent one in another, which we attribute to non-deterministic GPU kernels, so a seed here labels an independent run rather than a reproducible one. And it is regime-dependent: every divergence occurred in the fixed regime, while the randomised regime completed 5 of 5 runs. Randomising the training conditions appears to stabilise optimisation as well as to harden the codec, which we did not anticipate and do not test further.

7.3. The signal-to-noise axis is not where the problem is

Figure 7(a) and Table 6 sweep the test signal-to-noise ratio across 4.0 to 16.0 dB, a range chosen to contain the link-budget swing measured in Section 5 rather than picked for effect. A

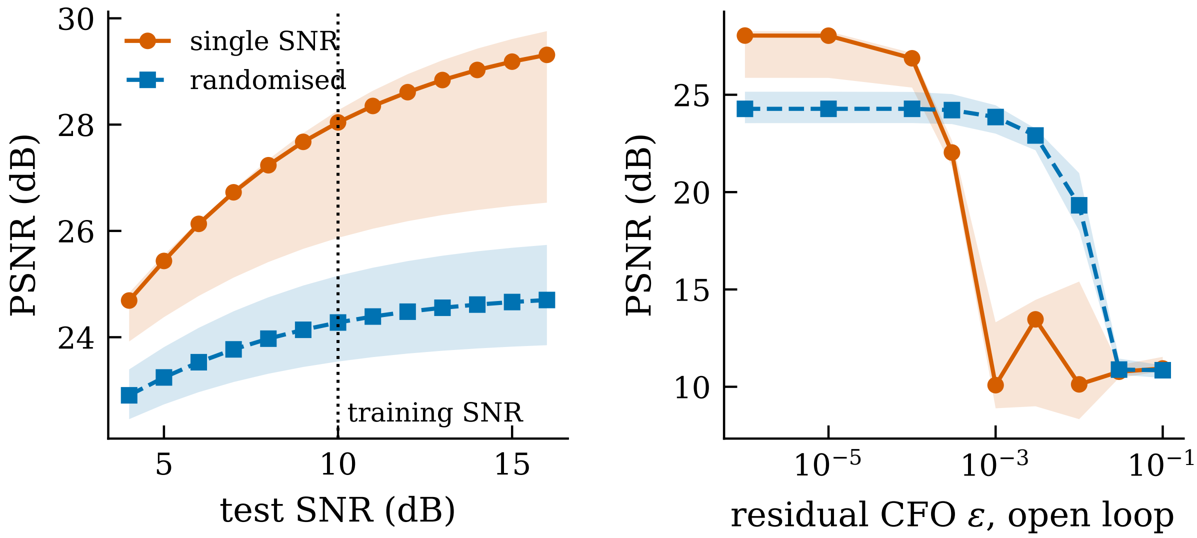

Figure 7: A codec trained at a single operating point, evaluated across the conditions Section 5 measured, over the 7 valid runs of the fixed regime and the 5 of the randomised regime. (a) test signal-to-noise ratio; (b) residual normalised frequency offset of (11) under the open-loop assumption, on a logarithmic axis with the zero-offset case at the left edge. "Single SNR" is the fixed-operating-point regime and "randomised" the regime that draws both quantities during training.

Table 6: Reconstruction quality in dB under the two training regimes, median over the valid runs with the range across them. Offsets are open loop, that is $L = k$. The evaluation conditions are those measured in Section 5, not chosen for this table.

	trained at one SNR ($n = 7$)		randomised training ($n = 5$)	
Evaluation condition	median	range	median	range
SNR = 10 dB (training point)	28.04	25.87–28.27	24.28	23.54–25.16
SNR = 4 dB (low end of pass)	24.69	23.92–24.83	22.91	22.46–23.39
SNR = 16 dB (high end of pass)	29.31	26.53–29.76	24.70	23.85–25.74
open-loop CFO $\varepsilon = 10^{-4}$	26.87	25.37–27.15	24.28	23.55–25.15
open-loop CFO $\varepsilon = 10^{-3}$	10.08	8.89–13.31	23.86	23.00–24.46
open-loop CFO $\varepsilon = 10^{-2}$	10.12	8.34–15.41	19.32	18.00–20.98

codec trained at a single operating point reaches 28.04 dB there and 24.69 dB at the low end, a loss of 3.35 dB, and the degradation is monotone and gradual with no threshold.

This is the second result in this paper that runs against the audit rather than for it, and it is consistent with the first. Section 5 found the geometric term moves slowly enough that a transmission block sees a static channel; this section finds that even the pass-scale sweep costs a codec only a few decibels. Training at a single signal-to-noise ratio is a defensible choice, and an audit that stopped at the channel model and called the fixed operating point a flaw would be wrong.

7.4. The residual frequency offset is where the problem is, and it is a cliff

Figure 7(b) tells a different story. Under the open-loop model of (11), in which the offset accumulates across the whole block, the same codec loses 1.17 dB at $\varepsilon = 10^{-4}$ and 17.96 dB at $\varepsilon = 10^{-3}$, falling from 28.04 dB to 10.08 dB. The transition is a cliff rather than a slope and it lies between two adjacent decades, so a single evaluation point on this axis says almost nothing about the neighbouring one.

The spread is informative, and it is not small. At the training point the valid runs span 26.46 to 28.23 dB. At $\varepsilon = 10^{-3}$ they span 8.48 to 12.51 dB. A quantity whose seed-to-seed range is four decibels cannot be reported as a point estimate, which is why Table 6 carries ranges throughout and why the comparisons below are made between medians whose ranges do not overlap.

7.5. What sets the cliff is the carrier re-estimation interval

The open-loop model is the strongest assumption in this paper, and it is worth attacking rather than defending. A receiver that re-estimates carrier phase periodically does not let the offset accumulate for a whole block, and if the collapse depends on accumulation then it should move, predictably, when the accumulation is interrupted.

We therefore keep the codec and the bandwidth ratio fixed and reset the phase every L symbols, so that (11) becomes $\varphi_m = 2\pi\varepsilon\,(m \bmod L)$, with $L = k$ the open-loop case and $L = 1$ perfect tracking. Changing L rather than k matters: k is tied to the bandwidth ratio through (8), so sweeping it would confound accumulation length with compression, while L leaves the codec untouched. Substituting L for k in (12) predicts a cliff at $\varepsilon = 1/(2(L-1))$, which across the 5 values of L tested moves by a factor of 256, four decades.

Figure 8 reports the outcome and it is the clearest result in the paper. With perfect tracking the codec is flat across the whole sweep, reaching 28.04 dB at $\varepsilon = 10^{-3}$ where the open-loop case has fallen to 10.08 dB. Re-estimating every 2 symbols is almost as good, 27.52 dB, and the intervals between fall in order, 28.04, 28.02 and 23.47 dB. Panel (b) places the measured cliff against the predicted one: the 5 points fall on a line parallel to the diagonal at a constant $0.25\times$, with a range of 0.19 to 0.42 across the factor of 256 in L.

The scatter in that constant is not evidence against the scaling, and its size is set by the measurement rather than by the codec. The offset grid steps by about $3.3\times$ per point, so the first point falling three decibels below the clean value locates the cliff only to within that spacing; a scatter of roughly a factor of two in the ratio is what a grid of that resolution produces. Refining the grid would tighten the constant without changing the slope, which is the quantity under test.

What that constant is, and is not, matters. Equation (12) predicts the *scaling* correctly: the cliff moves inversely with the accumulation length over the whole range tested. It does not predict the absolute position, and the offset has a plain reading rather than being an invariant we discovered. Since (12) is linear in ε, a measured cliff at 0.25 of the $\Delta\varphi = \pi$ criterion is simply the statement that three decibels are already lost by the time the tail of a block has rotated 0.25π, which is about 45°. The number therefore encodes the loss threshold we chose, not a property of the codec: a stricter threshold would move it down and a looser one up, while leaving the slope untouched. The scaling is the result; the constant is a unit conversion between a phase criterion and a decibel criterion.

What this does to the compensation figure. It qualifies it, and the qualification is the substantive one. A residual of $\varepsilon = 10^{-3}$ at a symbol rate of 10 Msym/s is 10 kHz, which against the median Ka-band shift of 359.2 kHz is ninety-seven per cent compensation. But that number is a statement about a

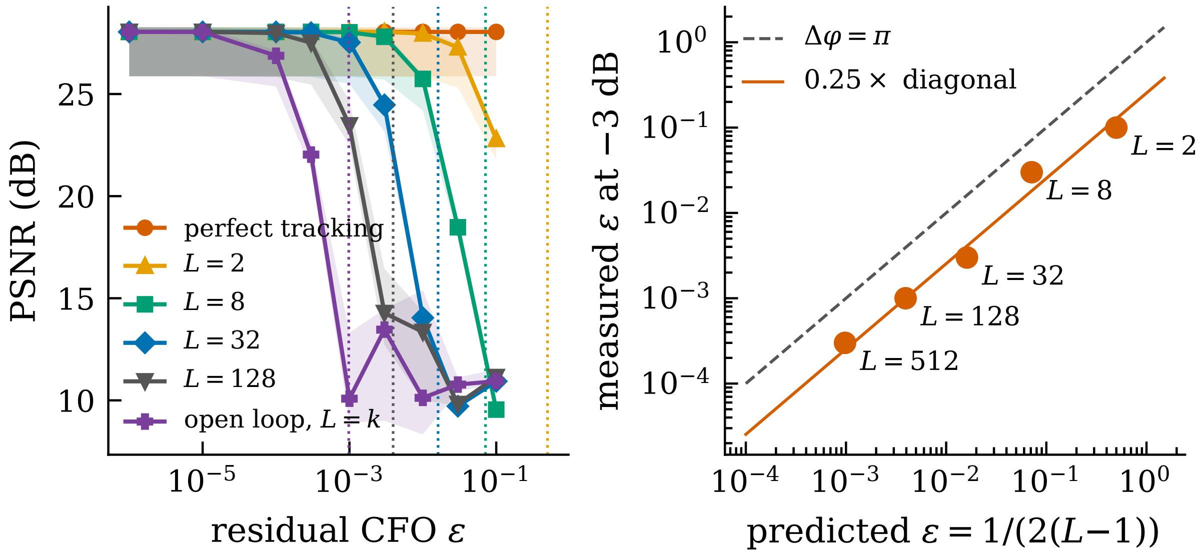

Figure 8: The collapse is governed by accumulated phase, not by Doppler as such. (a) quality against residual offset for four carrier re-estimation intervals, medians over the 7 valid runs of the fixed regime with bands spanning them, dotted verticals marking the position (12) predicts for each L. (b) measured cliff, taken as the first three-decibel loss, against the predicted position; the points lie parallel to the diagonal at $0.25\times$.

receiver that lets the residual accumulate over 512 symbols. The same receiver re-estimating every 2 symbols tolerates roughly 256 times the residual, and a tracking loop tighter still is unaffected at any offset tested here. The tolerable residual is therefore a property of the pair (codec, carrier recovery interval) and not of the codec alone, which is precisely why it belongs in the evaluation and precisely what no work in the population states.

7.6. A remedy, and the limits of it

Randomising the signal-to-noise ratio and the residual offset during training, over the ranges Section 5 measured, holds 23.86 dB at open-loop $\varepsilon = 10^{-3}$ against 10.08 dB for the fixed regime, a gain of 13.77 dB, at a cost of 3.76 dB at the design point.

Two qualifications keep this from being a general recommendation. The remedy is specific to the axis it targets: on the signal-to-noise sweep the randomised codec is *worse* at the low end by -1.78 dB, so randomising an axis that was not a problem spends capacity for nothing. And it moves the cliff rather than removing it, since at $\varepsilon = 10^{-2}$ the randomised codec has itself fallen to 19.32 dB. What the comparison establishes is narrower and more useful than a fix: the failure lives in the training distribution rather than in the architecture, and it is therefore visible to anyone who evaluates for it.

8. Recommendations

Three things follow, in decreasing order of confidence.

State the channel, and state it in the units of the orbit. An evaluation over additive white Gaussian noise at a fixed signal-to-noise ratio is a legitimate experiment and this paper does not argue otherwise. What is missing is the translation: at what elevation, at what point in a pass, and under what residual frequency offset does the stated operating point sit. The generator released here performs that translation from public orbital elements in a few lines.

Report the operating range, not the operating point. A visibility window of 206.0 s over which the geometric term alone moves 3.72 dB is the interval a deployed codec must serve. Reporting a single number at the centre of that interval describes the best case.

Treat residual frequency offset as a swept parameter. Doppler is compensated in practice and no serious system leaves 359.2 kHz uncorrected. What matters is the residual, and Section 7 shows why: between $\varepsilon = 10^{-4}$ and 10^{-3} the codec loses 17.96 dB under open-loop accumulation, so a single evaluation point on this axis carries almost no information about the neighbouring decade. State the carrier recovery interval alongside it, because Section 7.5 shows the two cannot be separated: the same codec is unharmed at the same offset when phase is re-estimated every 2 symbols. Sweeping both costs nothing beyond the sweep, and (12) says where to look.

9. Threats to validity

One survey is one sampling frame. The population is a convenience sample drawn from a single recent survey. It is not a systematic review, and a work absent from that survey is absent here. What the sample buys is that the pairing between limitation and work is the field's rather than ours, which is the specific claim under test.

Two fifths of the population could not be read. Full text was obtained for 27 of 46 works. The 19 that could not be obtained may well instantiate effects the audit does not credit them with. They are excluded from every count rather than assumed absent, and the denominator 26 is stated wherever a proportion appears.

The population mixes orbital and aerial settings. Relative velocity on an aerial link is orders of magnitude below orbital, so Doppler is not the binding constraint there and the aerial works should not be expected to instantiate it. If the two zeros existed only because 7 aerial and 4 integrated works were pooled with the satellite ones, the finding would be an artefact of the pooling. Restricting to the 15 satellite works alone, time-varying Doppler is instantiated by 0 of them and an orbital trace by 0, so both zeros survive the restriction unchanged. On the effects that are not zero the subgroup behaves as expected, with 9 of 15 evaluating over additive white Gaussian noise and 7 naming Doppler in their evaluation.

The reset model is optimistic, and in a known direction. Resetting the phase every L symbols in Section 7.5 is a perfect reset: the estimate is exact and costs nothing. A real carrier loop estimates from noisy symbols, lags a changing offset and consumes pilot overhead, so it will do worse than the reset at the same interval. The direction of that bias is therefore known: the reset curves are an upper bound on what tracking buys, and the open-loop curve is the lower bound. The true behaviour of a given receiver lies between two curves the paper reports rather than outside them, which is the reason for reporting both ends rather than one estimate in the middle.

The channel model is open loop by default. Equation (11) lets a residual offset accumulate for a whole block unless L says otherwise, which is one end of a range of receiver designs rather than a neutral choice. Section 7.5 treats it as a swept parameter for that reason, and the headline compensation figure is stated with the interval it assumes. What the paper does not model is a closed-loop estimator with its own noise, whose tolerable residual would be set by loop bandwidth rather than by a reset interval.

Absence of a phrase is not absence of a model. The census detects the vocabulary in which an effect is normally described. A work could model time-varying Doppler without using any of the phrasings the controls test. This is why the coding sheet is released with the sentences attached rather than only the counts.

The orbital measurement is one constellation. All figures come from Starlink elements at one epoch. Other shells at other altitudes will shift the numbers, though the geometry that produces them does not change in kind.

Free-space path loss is a lower bound, not the link budget. The swing reported here excludes antenna pattern, atmospherics and noise figure. Every one of them widens the range, so the direction of the bound is known even though its tightness is not.

10. Conclusion

Surveys of semantic communication for non-terrestrial networks map each constraint of the satellite link onto the works said to address it. Measuring both sides of that mapping shows where it is load-bearing and where it is not. The orbit produces a Doppler shift with a median peak of 359.2 kHz drifting at up to 16.9 kHz/s and a link budget that sweeps 3.72 dB across a 206.0 s window. Of the 26 works of Section 6 that could be read and coded, 16 evaluate over additive white Gaussian noise, 4 use a standardised non-terrestrial model, and none instantiates either the drift or the geometry that produces it.

Applying the measured conditions to a codec trained under the modal assumption then locates the consequence. It is not on the axis an audit would reach for first: the pass-scale signal-to-noise sweep costs 3.35 dB and degrades gracefully. It is on the axis nobody instantiates, where a residual frequency offset of 10^{-3} costs 17.96 dB once it is allowed to accumulate across a block. That qualification is the result rather than a caveat on it: the cliff moves inversely with the accumulation length across a factor of 256, at a constant 0.25 of what the accumulated-phase argument predicts, and vanishes under carrier tracking. The quantity a deployed system must respect is not Doppler but uncorrected accumulated phase, and it is fixed by the codec and the receiver together.

The reading is not that this work is unsound, nor that its channel models are careless. It is that the field's evaluations are strongest on the effect that turns out not to matter and silent on the one that does, and that closing the distance is cheap: the orbital elements are public, the propagation is a library call, and the harness that connects them to an existing codec is released with this paper.

Acknowledgment

The author used a large language model (Anthropic Claude) as an assistant while preparing this work: for drafting and revising prose, for writing and debugging the analysis and figure- generation code, and for reviewing intermediate drafts. Every experimental design decision, every claim, and every number reported here was checked by the author against the released result files, and the released repository records the assistance in its commit history. The tool is not an author and bears no responsibility for the content.

Data and code availability

The checklist, the coding sheet with the sentence justifying every decision, the orbital generator, the detector controls and the evaluation harness are released at

`https://github.com/haodpsut/ntn-semcom-realism-audit`

The repository also carries the raw result files rather than summaries of them, so every number in this paper can be recomputed from the command that produced it.

Where each measurement ran. A paper that audits others for not stating the conditions of their evaluation should state its own. The orbital measurements of Section 5, the census of Section 6 and all figure and table generation ran on a workstation (Apple M5, Python 3.9.6, NumPy 2.0.2, SGP4 2.25). The codec experiments of Section 7 ran on a server with an Intel Xeon Gold 6148 and a NVIDIA GeForce RTX 4090 (Python 3.11.16, PyTorch 2.5.1+cu121, CUDA 12.1).

Splitting the work across two machines is worth declaring because floating-point results are not always portable between them. No measurement in this paper ran on both, so no reported quantity has two values, and no number produced on one machine is compared directly against a number produced on the other: the orbital side is pure geometry on the workstation and the codec side is training on the server. Where the two meet, in Section 7, the orbital results enter only as the endpoints of a sweep.

Appendix A. The population, enumerated

Table A.7 lists every work in the sampling frame. It exists so that the census can be checked rather than taken on trust: a reader who disagrees with a coding decision can find the work, and a reader who wants to know what the audit could not see can find that too.

Table A.7: All 46 works in the sampling frame. "Coded" means full text was retrieved and an evaluation section could be located; "no eval. section" means the text was retrieved but carries no conventional evaluation; "not retrieved" means no full text could be obtained, and those works are excluded from every count rather than assumed absent.

Ref. in [1]	This paper	Status	Title
[81]	[14]	coded	Joint semantic coding and routing for multi-hop semantic t
[93]	[12]	coded	Semantic satellite communications based on generative foun
[94]	[11]	coded	Importance-aware robust semantic transmission for LEO sate
[95]	[8]	coded	Semantic enabled 6G LEO satellite communication for earth
[100]	[37]	coded	Semantic satellite communications for synchronized audiovi
[102]	[20]	coded	Knowledge graph driven UAV cognitive semantic communicatio
[104]	[24]	coded	SemSpaceFL: a collaborative hierarchical federated learnin
[108]	[32]	no eval. section	Goal-oriented communications for interplanetary and non-te
[109]	—	not retrieved	Channel code-book (CCB): semantic image-adaptive transmiss
[110]	—	not retrieved	Semantics-aware updates from remote IoT devices to interco

Ref. in [1]	This paper	Status	Title
[112]	[28]	coded	Deep reinforcement learning-based resource allocation for
[113]	[9]	coded	Compressed learning-based on-board semantic compression for
[114]	[29]	coded	Energy-efficient probabilistic semantic communication over
[115]	[17]	coded	Semantic communication enabled 6G-NTN framework: a novel d
[116]	[31]	coded	Semantic-forward relaying: a novel framework towards 6G co
[117]	[15]	coded	Semantic forwarding and codebook-enhanced model division m
[118]	[25]	coded	Semantic-aware task clustering for federated cooperative m
[137]	—	not retrieved	Doppler-adaptive digital semantic communication for low ea
[154]	[10]	coded	Cognitive semantic augmentation LEO satellite networks for
[161]	[13]	coded	Generative ai agents with large language model for satelli
[162]	—	not retrieved	Generative semantic communication for efficient utilizatio
[164]	—	not retrieved	Semantic communication-aware end-to-end routing in large-s
[165]	[16]	coded	Semantic transmission framework in direct satellite commun
[167]	—	not retrieved	Energy efficient probabilistic semantic communication over
[168]	—	not retrieved	QoSem-based resource allocation for semantic communication
[170]	[30]	coded	A joint jscc-resource allocation framework for qos-aware s
[171]	—	not retrieved	Semantic communication in satellite-borne edge cloud netwo
[174]	—	not retrieved	Semantics-empowered UAV-assisted wireless communication sy
[178]	[18]	coded	Robust semantic transmission for low-altitude UAVs: predic
[179]	[21]	coded	Context-aware information transfer via digital semantic co
[180]	[36]	coded	Personalized saliency in task-oriented semantic communicat
[182]	—	not retrieved	Resource allocation and intelligent trajectory optimizatio
[184]	—	not retrieved	NOMA-based intelligent resource allocation and trajectory
[186]	[23]	coded	Semantic-aware UAV command and control for efficient IoT d

Ref. in [1]	This paper	Status	Title
[190]	[19]	coded	Low-altitude multi-uav-assisted data collection and semant
[191]	[22]	coded	Fair resource allocation in uav-based semantic communicati
[196]	—	not retrieved	Federated learning semantic communication in UAV systems:
[198]	—	not retrieved	Resource allocation for semantic communication in ntn down
[199]	—	not retrieved	Attention–based haps–to–ground nodes optimization for diff
[200]	[26]	coded	Twinning for space-air-ground-sea integrated networks: bey
[202]	—	not retrieved	Perception-aware offloading with collaborative ground–spac
[203]	[27]	coded	Aerial semantic relay-enabled SAGIN: joint UAV deployment
[204]	—	not retrieved	Semantic-aware resource allocation in mec-assisted sagin:
[205]	—	not retrieved	Semantic communication for high-speed mobility scenarios:
[206]	—	not retrieved	Generative ai-aided vertical handover decision in sagin fo
[207]	—	not retrieved	Satellite-aided low-altitude uav service migration with se

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